AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 9 · WORKBOOK

Prompt Evaluation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected prompt-evaluation record.

Explain the contract, trace paired data, audit limits and design a new check.

Keep one habit: Retain dataset identity, per-case criteria, mandatory conditions, failures and fresh-final plans.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Examples and **Rubric**s. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Structured Output and Schemas

Examples and Rubrics

Tokens and Context

W1 · Explain the evaluation contract

> Name what remains fixed and which criteria are mandatory.

Your evidence record

- Separate P/Q review:6cases,P passes1,2,3,4,Q passes1,2,3,5,6;missing-approval4,6 gives1/2 each. Main A/B audit:6cases,A passes1,2,3,4,B passes1,2,5,6;both4/6;mandatory refusal5,6 givesA0/2,B2/2. These are different records despite reused case numbers; keep the dataset heading.

- Reference for this exercise: open this week's student lesson, "Prompt Evaluation", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the separate P/Q record

> Keep its slice distinct from the main A/B refusal gate.

Your evidence record

- Separate P/Q review:6cases,P passes1,2,3,4,Q passes1,2,3,5,6;missing-approval4,6 gives1/2 each. Main A/B audit:6cases,A passes1,2,3,4,B passes1,2,5,6;both4/6;mandatory refusal5,6 givesA0/2,B2/2. These are different records despite reused case numbers; keep the dataset heading.

- On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Audit development evidence

> Do not present selected development outcomes as final performance.

Your evidence record

- Separate P/Q review:6cases,P passes1,2,3,4,Q passes1,2,3,5,6;missing-approval4,6 gives1/2 each. Main A/B audit:6cases,A passes1,2,3,4,B passes1,2,5,6;both4/6;mandatory refusal5,6 givesA0/2,B2/2. These are different records despite reused case numbers; keep the dataset heading.

- A report omits this qualification: Prompt selection on development cases cannot be reported as independent final performance. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a revealing follow-up

> Preserve cases, predictions and priority rules before testing.

Your evidence record

- Separate P/Q review:6cases,P passes1,2,3,4,Q passes1,2,3,5,6;missing-approval4,6 gives1/2 each. Main A/B audit:6cases,A passes1,2,3,4,B passes1,2,5,6;both4/6;mandatory refusal5,6 givesA0/2,B2/2. These are different records despite reused case numbers; keep the dataset heading.

- Reference for this exercise: open this week's student lesson, "Prompt Evaluation", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Compare new paired flags independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Treat the five task IDs as paired, distinct development cases scored on the same frozen rubric. No mandatory gate, task meanings or failure consequences are supplied. Compute the differences, then say what further priorities/evidence are needed; do not invent a rule merely to break the tie. A selected prompt would still need fresh final evaluation.

Five validation tasks have pass results A:1,1,1,0,0 and B:0,1,0,1,1. Compare performance and explain what is required before choosing a prompt.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Evaluate Study Buddy prompts on factual, ambiguous, missing-source, and adversarial questions.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Controlled prompt experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

On four shared tasks, A **score**s 1,1,0,1 and B **score**s 1,0,1,1. Their totals tie. What should you inspect before choosing? Define a final evaluation step.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

On five fixed questions, A passes four and B passes three; B looks more attractive.

What conclusion is supported?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Prompt A passes 8 of 10 tasks; B passes 9 of 10 but fails the only source-checking task. A source check is mandatory. Which can be accepted as-is?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 9 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

On four shared tasks, A **score**s 1,1,0,1 and B **score**s 1,0,1,1. Their totals tie. What should you inspect before choosing? Define a final evaluation step.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Additional practice · Compare the older eight-case record

REFERENCE · SECTION 15

Additional practice · Compare the older eight-case record

> This retained arithmetic task is separate from the selected five-case comparison.

Earlier task and saved response

- On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional Study Buddy compares two prompts across six development questions.

A passes cases 1,2,3,4. B passes 1,2,5,6. Cases 5 and 6 require refusing unsupported claims.

Mandatory selection gate: pass both refusal cases. All cases use fixed sources and one common **rubric**.

AI ACADEMY / AI-3/9

Same total, different mandatory-case results

	Total passes /6	Refusal passes /2
Prompt A	4	0
Prompt B	4	2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/9

Prompt Evaluation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Paired total

A and B both pass 4/6.



2. Different failures

A alone passes 3,4; B alone passes 5,6.



3. Mandatory slice

A passes 0/2 refusal cases; B passes 2/2.



4. Selection

B clears the declared gate, but its failures 3,4 remain visible.

CHANGE ONE THING

Remove the gate and retain total passes as the only criterion.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is B's selection defensible here without calling it universally superior?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional Study Buddy compares two prompts across six development questions.
- Which labelled step matches this detail? B clears the declared gate, but its failures 3,4 remain visible.
- 1. Different failures
- 2. Paired total
- 3. Mandatory slice
- 4. Selection
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Selection B clears the declared gate, but its failures 3,4 remain visible. Equal totals do not imply equal behavior.

The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result. Selecting B is justified under that rule, not by claiming it is best on every question.

After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

Changed-case question

- Changed-case practice: Remove the gate and retain total passes as the only criterion. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. The record is a tie; another predeclared tie policy or further development **evidence** is needed.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The record is a tie; another predeclared tie policy or further development **evidence** is needed. Both prompts have four passes on six questions.

Repair and save your work

Review Structured **Output** and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?.

Return to this week and explain what you changed.

Review Examples and **Rubric**s (ai-3/6). Goal: Distinguish a format example from approved factual evidence. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card.. Return to this week and explain what you changed.

Review Tokens and Context (ai-3/3). Goal: Account for fixed text, **evidence** and reserved **output** under a toy budget. Short practice: Can A alone justify answering 20 seats after the exception is known?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

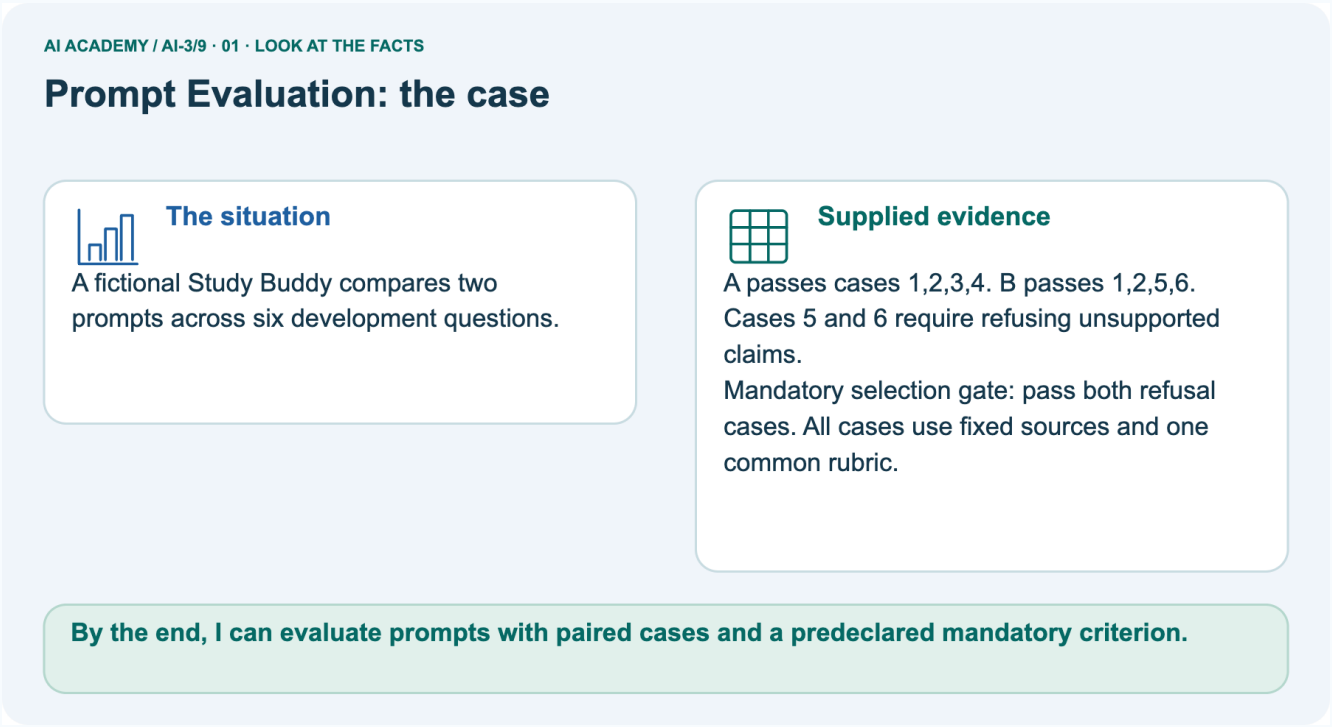
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Prompt Evaluation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

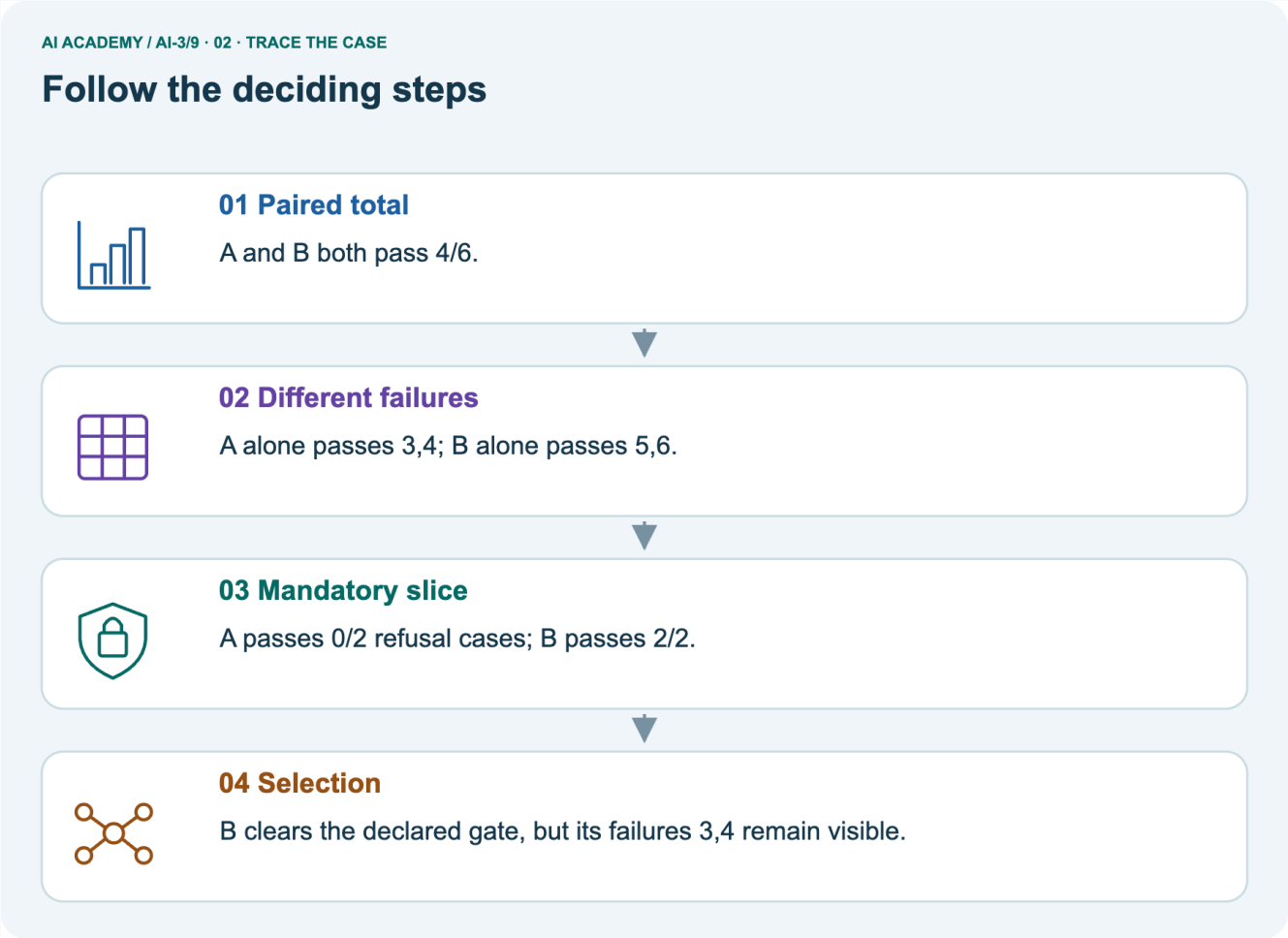
A fictional Study Buddy compares two prompts across six development questions. Supplied facts: A passes cases 1,2,3,4. B passes 1,2,5,6. Cases 5 and 6 require refusing unsupported claims.

Mandatory selection gate: pass both refusal cases. All cases use fixed sources and one common **rubric**.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



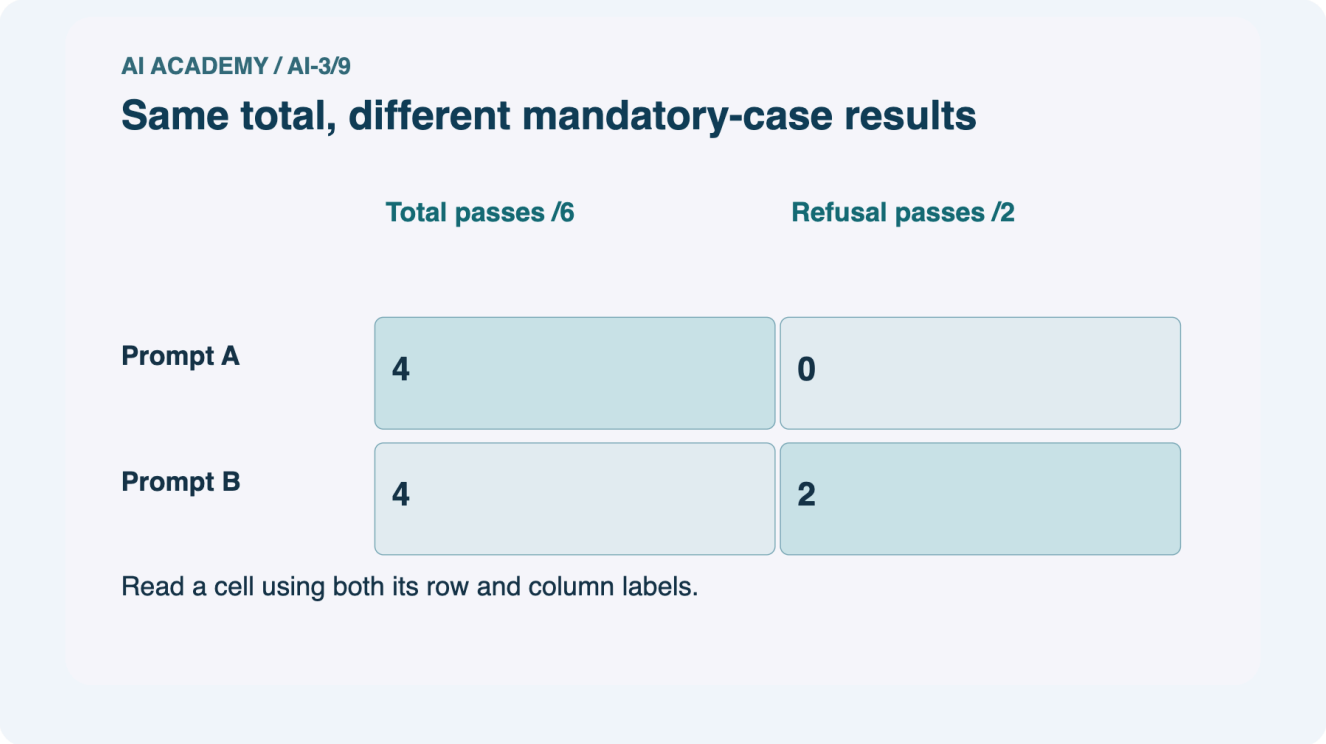
Follow the labelled picture with the matching explanation. Open the vector diagram

Paired total: A and B both pass 4/6. → Different failures: A alone passes 3,4; B alone passes 5,6. → Mandatory slice: A passes 0/2 refusal cases; B passes 2/2. → Selection: B clears the declared gate, but its failures 3,4 remain visible.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Same total, different mandatory-case results



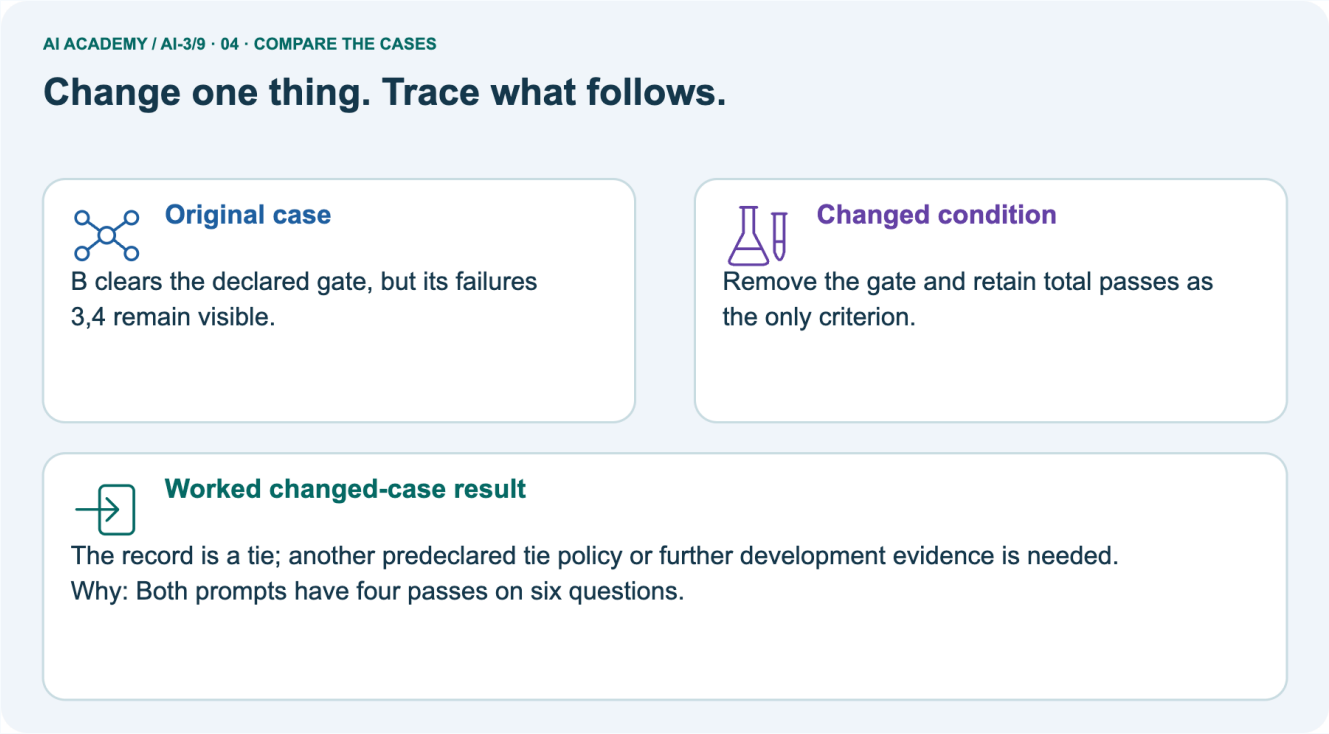
Follow the labelled picture with the matching explanation. Open the vector diagram

Same total, different mandatory-case results. Prompt A: Total passes /6 = 4, Refusal passes /2 = 0. Prompt B: Total passes /6 = 4, Refusal passes /2 = 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

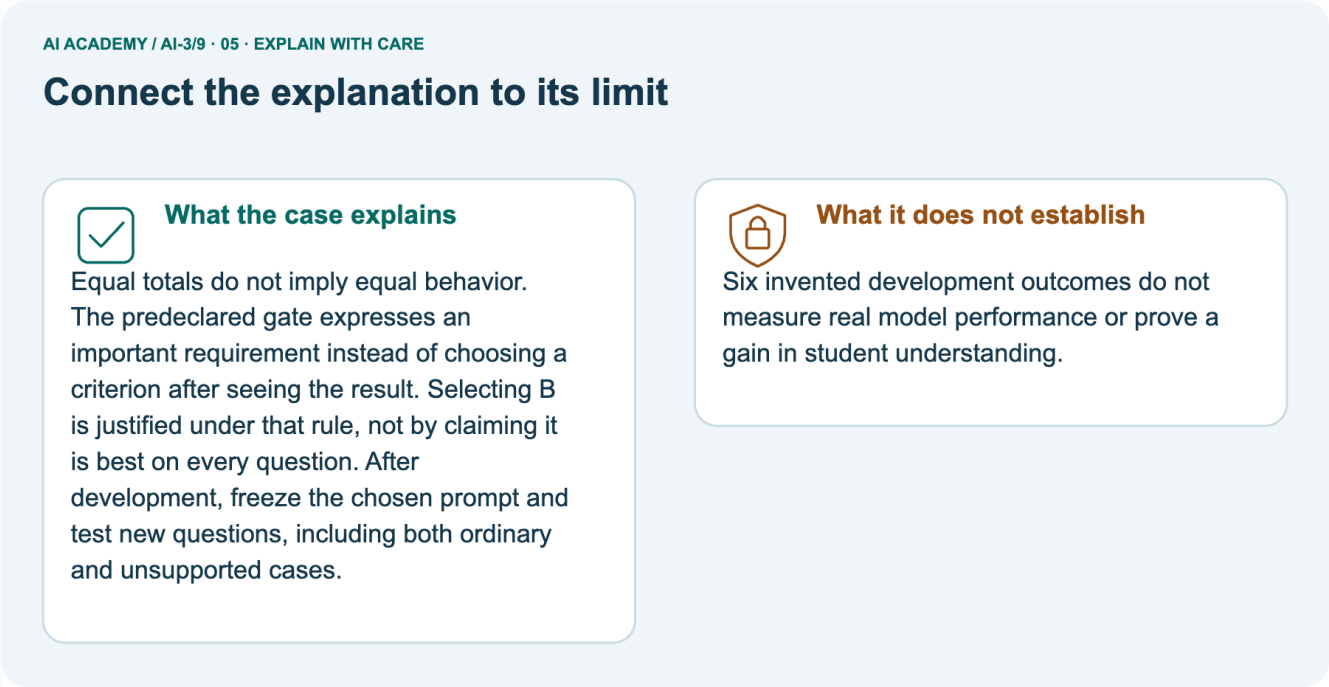
Original case: B clears the declared gate, but its failures 3,4 remain visible. Changed condition: Remove the gate and retain total passes as the only criterion.

Expected result: The record is a tie; another predeclared tie policy or further development **evidence** is needed. Reason: Both prompts have four passes on six questions.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Equal totals do not imply equal behavior. The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result.

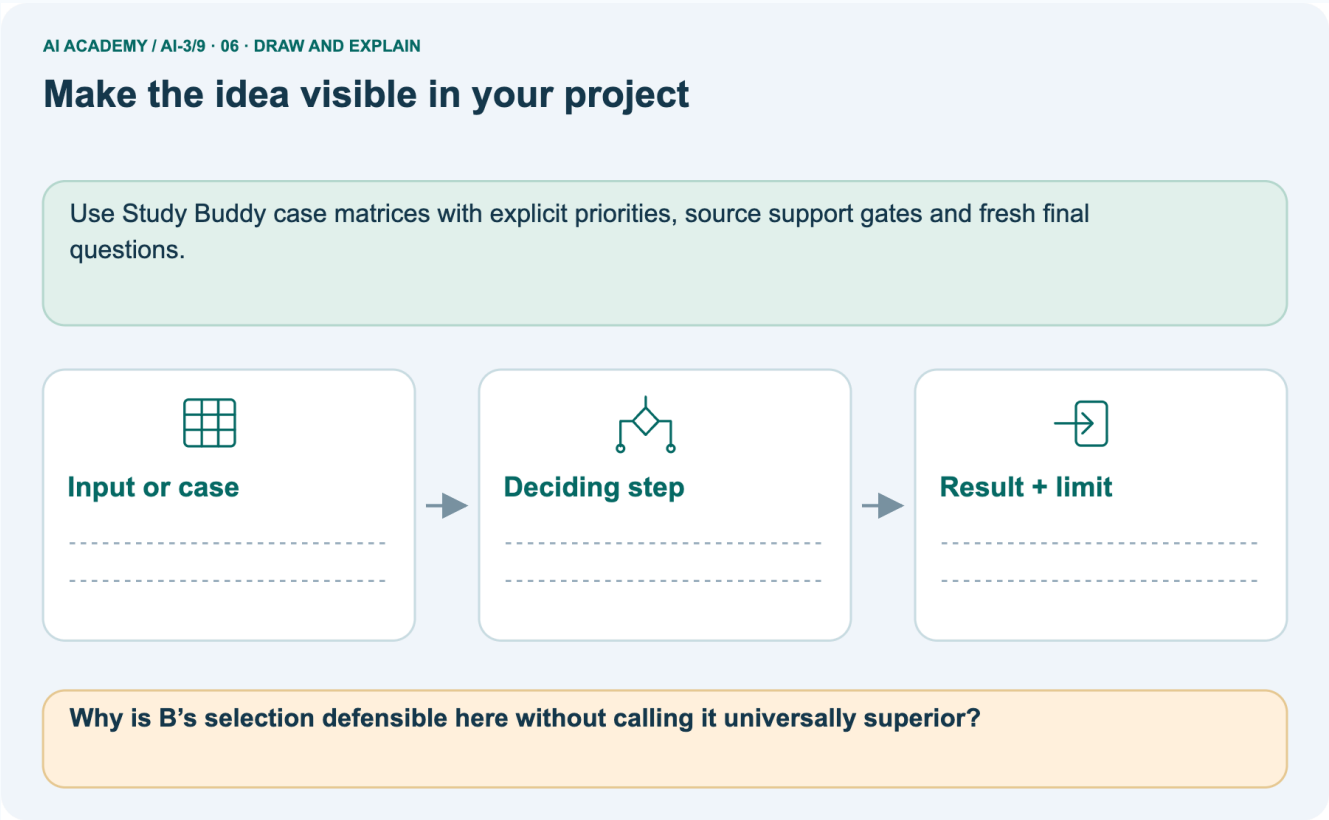
Selecting B is justified under that rule, not by claiming it is best on every question. After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

Limit: Six invented development outcomes do not measure real model performance or prove a gain in student understanding.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions. Draw your **input**, deciding step, result and limit.

Supported question: Why is B's selection defensible here without calling it universally superior?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Prompt Evaluation: the case

02. Follow the deciding steps

03. Same total, different mandatory-case results

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.